

Wael Bahgat

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Professional Summary

Principal Software Engineer with 15+ years of experience building scalable, high-performance systems in telecom, aerospace, and satellite communications. Expert in C++17, object-oriented design, and architecture using SOLID principles. Demonstrated leadership across full lifecycle delivery including CI/CD pipeline design, container lifecycle management, multi-architecture deployments, and production debugging.

Core Competencies

- **Languages & Frameworks:** C++17, C++, Java, Flutter, Qt, SQL, Python (basic)
- **DevOps & Tools:** GitLab CI/CD, Docker, AWS (ECS, Fargate, RDS, S3, CloudFront, ALB), GTest, GCov, Git, aarch64 cross-compilation, systemd
- **Architecture:** Modular design, plugin architecture, multithreading, REST APIs, SOLID principles, design patterns
- **Testing:** Unit Testing (GTest), TDD, GCov coverage analysis, performance/load testing
- **Databases:** PostgreSQL, MongoDB, SQLite3, Oracle, MySQL
- **Other:** Release engineering, CI/CD pipelines, production debugging, performance profiling, cross-functional team collaboration

Professional Experience

Principal Software Engineer (Lead Developer, Core Network Team)

Lynk Global — Dec 2023 – Present

- Architected dual-mode (satellite / ground) GSM/LTE core network in C++17: HLR, MME, HSS, DRA, MSC, and LynkDBGw — including GSUP protocol encoding/decoding
- Designed modular plugin-based framework with SOLID principles, hot-reload JSON configuration, and runtime-swappable database backends (MongoDB, PostgreSQL, SQLite3)
- Built Docker orchestration for multi-stage satellite and ground deployments, optimizing layer reuse and enabling targeted component updates
- Achieved horizontal scalability through centralized PostgreSQL and stateless network nodes; vertical scaling via async thread-pool processing
- Designed and owns a 4-mode GitLab CI/CD release pipeline: nightly unit tests, two-phase Docker builds, RC creation with JSON manifest tracking, and final amd64/aarch64 multi-arch tagging
- Built container lifecycle infrastructure: lynk-executor runtime manager (4 deployment modes), Docker registry management, and systemd production integration
- Integrated GTest unit testing and GCov coverage analysis across all components

Senior Software Specialist

L3Harris — Sep 2021 – Dec 2023

- Built core components for aviation surveillance systems using C++.
- Designed and implemented a framework to manage multiple service applications sharing a common core with extendable layers tailored to each application.
- Developed key components including a configuration manager, log manager, thread pool, and a pluggable business logic layer.
- Integrated the system with GTest for unit testing and GCov for code coverage analysis.
- Implemented multithreaded services with XML-driven configuration.
- Employed TDD and performance testing strategies with GTest and custom profilers.

Development Team Lead (Lead Developer)

Gibbs & Cox (A Leidos Company) — Jul 2020 – Sep 2021

- Led development of an autonomous navigation system for surface vessels.
- Designed a modular Qt-based software framework leveraging object-oriented design and software patterns.
- Applied the Composite design pattern to model onboard devices hierarchically, enabling dynamic and scalable system composition.

- Designed a platform supporting vessel configuration flexibility, hardware abstraction, and runtime extensibility.
- Directed full software lifecycle: requirements, architecture, implementation, testing, and deployment.
- Facilitated daily standups and sprint planning to drive team efficiency and timely delivery.

Senior Software Specialist

L3Harris — May 2016 – Jul 2020

- Aggregated radar and ADS-B data in real time across multiple sites.
- Built drone surveillance UI in C++/Qt.
- Worked on VOIP air traffic control communications.
- Developed reusable C++ libraries with GTest coverage.
- Part of the team developing a voice switching solution for air traffic control allowing hybrid analog and VOIP calls based on configuration.

Lead Software Engineer – Production Engineering

Openet Telecom — May 2012 – Apr 2016

- Delivered hotfixes and patches across the production stack in C++ and Java.
- Analyzed memory leaks, deadlocks, and crashes using gdb and mdb.
- Coordinated with global clients and internal teams to resolve critical issues.

Lead Software Engineer – Professional Services

Openet Telecom — Jan 2011 – May 2012

Senior Software Engineer → Software Engineer

Openet Telecom — Jan 2005 – Dec 2010

- Designed full-stack telecom billing solutions for major carriers.
- Developed using DSD, Java, and PL/SQL in high-volume Oracle DB environments.
- Delivered projects for Verizon, AT&T, Sprint, and others.
- Authored requirements, performance test specs, and architecture documents.

Projects

Yalabeena — Mobile & Web Session Booking Platform (formerly BMyGst)

yalabeena.com

- Built a full-stack session booking platform using Flutter (web/mobile frontend) and C++17 Drogon REST backend
- Integrated REST APIs, Braintree payments, Google Calendar, JWT sessions, credit ledger, and email notifications
- Deployed on AWS ECS Fargate with RDS PostgreSQL and S3 hosting

Education

M.S. in Electrical Engineering

NYU Tandon School of Engineering (formerly Polytechnic University), Brooklyn, NY

GPA: 3.87 | Graduated: Jan 2008

B.S. in Computer Engineering

Ain Shams University, Cairo, Egypt

Graduated: Jan 2001